Technology Stack

Date	30 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*React.js, HTML5 , CSS3, JavaScript, Vite*	*Provides a responsive, component-based user interface for entering soil parameters and displaying crop recommendations.*

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2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python, Flask*	*Processes user requests, loads the trained machine learning model, and returns crop predictions through REST APIs .*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*CSV Dataset, Joblib (.pkl Model)*	*Stores the agricultural dataset used for training and the trained Random Forest model used during prediction.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Local Flask Server & Vite Development Server*	*Used to develop, test, and run the application locally before deployment.*
5	**Version Control & CI/CD**	*Git, GitHub*	*Tracks source code changes, enables collaboration, and stores the project repository.*

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Axios, Jupyter Notebook*	*Used for machine learning, data preprocessing, visualization, API communication, and model development.*